

LAKE CHARLES NWS, LA, USA

WMO: 722400

Lat: 30.1253N

Lon: 93.2161W

Elev: 6

StdP: 101.25

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03937

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -0.9	(c) 0.9	(d) -8.5	(e) 1.8	(f) 3.9	(g) -5.7	(h) 2.3	(i) 6.5	(j) 10.5	(k) 11.3	(l) 9.3	(m) 11.2	(n) 3.0	(o) 20	(p) 0.404

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.4	(c) 35.0	(d) 25.4	(e) 34.0	(f) 25.4	(g) 33.1	(h) 25.3	(i) 27.4	(j) 31.4	(k) 26.9	(l) 30.9	(m) 26.6	(n) 30.6	(o) 3.2	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.3	21.8	29.0	25.9	21.2	28.7	25.5	20.8	28.5	86.2	31.4	84.4	31.0	82.6	30.5	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 9.0	(b) 8.1	(c) 7.3	(e) -3.8	(f) 37.0	(g) 1.9	(h) 1.5	(i) -5.1	(j) 38.1	(k) -6.2	(l) 39.0	(m) -7.3	(n) 39.8	(o) -8.7	(p) 40.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.7	11.4	13.5	16.8	20.3	24.5	27.5	28.4	28.6	26.4	21.5	16.3	12.6
	DBStd	7.22	5.39	5.17	4.49	3.55	2.63	1.63	1.39	1.61	2.41	3.92	4.60	5.19
	HDD10.0	113	48	22	6	0	0	0	0	0	0	0	6	32
	HDD18.3	771	221	148	81	21	1	0	0	0	0	17	92	190
	CDD10.0	4017	92	121	218	311	448	526	572	576	492	357	194	112
	CDD18.3	1634	7	13	35	82	192	276	313	317	242	116	30	12
	CDH23.3	15563	9	22	130	495	1635	2891	3508	3506	2255	912	166	34
	CDH26.7	6136	0	0	7	63	506	1226	1545	1588	923	263	14	0
Wind	WSAvg	3.2	3.8	3.9	3.8	3.8	3.3	2.7	2.2	2.2	2.7	3.0	3.3	3.7
Precipitation	PrecAvg	1458	132	86	91	95	140	151	138	146	144	108	107	124
	PrecMax	2166	363	204	235	335	526	643	335	599	507	545	358	337
	PrecMin	884	11	11	5	3	1	21	12	0	11	0	3	51
	PrecStd	291	77	52	53	76	100	107	79	107	95	109	65	66
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.5	25.5	27.6	29.7	33.3	35.7	35.7	36.6	35.8	32.4	28.3	26.0
		MCWB	20.1	20.3	20.2	21.5	23.6	24.9	25.6	25.5	24.5	23.8	22.1	21.7
	2%	DB	23.0	23.9	26.1	28.3	31.5	33.8	34.4	35.1	33.8	30.8	26.7	24.3
		MCWB	19.1	19.9	20.5	21.7	23.4	24.8	25.7	25.7	24.6	23.3	21.5	21.0
	5%	DB	21.5	22.7	24.9	27.3	30.5	32.8	33.5	34.1	32.5	29.5	25.3	22.7
		MCWB	18.7	19.3	19.9	21.1	23.5	24.9	25.7	25.8	24.4	22.9	20.6	20.0
	10%	DB	20.0	21.3	23.5	26.2	29.5	31.9	32.7	33.0	31.4	28.1	23.9	21.1
		MCWB	17.7	18.6	19.3	20.7	23.3	25.0	25.7	25.7	24.2	22.1	20.0	18.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.6	22.2	22.7	24.5	26.7	27.5	28.0	28.0	27.5	26.4	24.3	23.2
		MCDB	23.2	24.0	25.1	26.8	29.3	30.7	32.3	32.2	31.1	29.3	26.2	24.7
	2%	WB	20.6	21.3	21.9	23.6	25.6	26.9	27.2	27.4	26.6	25.5	23.2	21.8
		MCDB	22.1	22.9	24.1	26.2	28.8	30.5	31.5	31.6	30.3	28.5	25.0	23.3
	5%	WB	19.6	20.4	21.3	22.9	24.9	26.4	26.8	26.9	26.0	24.7	22.2	20.8
		MCDB	20.9	21.9	23.5	25.5	28.3	30.2	31.0	31.1	29.7	27.6	24.2	22.4
	10%	WB	18.3	19.4	20.6	22.3	24.4	25.9	26.4	26.6	25.5	23.9	20.9	19.3
		MCDB	19.7	20.8	22.7	24.9	27.8	29.8	30.6	30.7	29.1	26.6	23.3	20.7
Mean Daily Temperature Range	5% DB	MDBR	10.3	10.0	10.3	10.2	9.4	8.6	8.4	8.9	9.7	11.1	11.1	10.3
		MCDBR	10.3	9.5	9.6	10.0	9.8	9.7	9.5	10.2	10.7	11.0	11.3	10.4
		MCWBR	7.4	6.2	5.2	4.7	3.8	3.2	3.1	3.1	3.7	4.8	6.6	7.3
	5% WB	MCDBR	8.8	7.7	8.1	7.9	8.2	8.0	8.6	8.9	8.6	8.6	9.6	9.2
		MCWBR	7.4	5.9	5.4	4.8	3.7	3.2	3.2	3.3	3.5	4.9	7.1	7.4
Clear-Sky Solar Irradiance	taub	0.342	0.358	0.376	0.431	0.466	0.500	0.500	0.488	0.458	0.382	0.367	0.352	
	taud	2.511	2.476	2.419	2.263	2.201	2.141	2.182	2.218	2.268	2.463	2.449	2.484	
	Ebn at Noon	891	906	910	866	834	800	799	806	819	868	853	859	
	Edh at Noon	89	100	113	137	147	155	148	142	130	101	94	88	
All-Sky Solar Radiation	RadAvg	2.87	3.28	4.54	5.50	6.14	6.14	5.91	5.60	5.01	4.41	3.32	2.55	
	RadStd	0.28	0.53	0.41	0.41	0.40	0.47	0.46	0.51	0.49	0.55	0.32	0.28	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.35	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+115	+79
(47) Regional (27 neighbors)		+0.30	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+110	+77

Nomenclature: See separate page